

Mr. Abdelhak BOURDIM

Embedded software engineer

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Fields of expertise:

- ◆ Requirements analysis and management
- ◆ **Embedded software** development: low level drivers, libraries and applications
- ◆ Applications development in C/C++
- ◆ Design review, debugging and code optimizations
- ◆ Evolutions monitoring and corrections
- ◆ Feasibility and POC
- ◆ Unit testing, continuous integration & End to End tests (IVV)
- ◆ Plans validation
- ◆ Documentation

Fields of intervention:

- ◆ Railway transport
- ◆ Aerospace
- ◆ Smart Grids (Energy)
- ◆ Medical systems
- ◆ Payment systems
- ◆ Access control

Technical fields:

- ◆ Operating systems: Windows, Linux
- ◆ IDE: Eclipse, VxWorks
- ◆ Embedded Tools: MPLAB, Code composer Studio (TI), IAR workbench, Keil µVision, LPCXpresso, Arduino, Raspberry pi
- ◆ Core architecture: MICROCHIP, ARM7, ARM9, CORTEX M0, CORTEX M3, CORTEX A8, Freescale, HITACHI H8S, NEC V850, TI DSP320, Infineon XE167HF, TI Concerto family.
- ◆ Programming languages: C, C++, C#, Java, Assembly, Bash, Perl, Awk, Python.
- ◆ Communication interfaces: RS232, SPI, I2C, RS485, PWM_ADC, CAN, LIN, USB, Ethernet
- ◆ Stacks: TCP/IP, USB, File Systems, Bluetooth BLE, Infrared (IRDA), RFID
- ◆ Embedded OS: UCOSII, FreeRtos, Linux, VxWorks
- ◆ Configuration management: Git, Clearcase, Synergy
- ◆ Development process: V-Model, Agile (JIRA)

Professional experience

Safran – Créteil

<July 2024 – May 2025>

Fields of expertise: Low-level driver update

Intervention title: SSPC

- **Goals and achievements:**

- ◆ Development in C
- ◆ Testing
- ◆ logging tools

- **Technical fields:**

C, Lauterbach TRACE32, Bash, Awk

Asterios Technologies – Massy

<Aug. 2023 – Feb. 2024>

Fields of expertise: Writing test procedures

Intervention title: T1042SOM – Software Testing

- **Goals and achievements:**

- ◆ Development in C
- ◆ Errata analysis
- ◆ Documentation

- **Technical fields:**

C, Bash, Python, Lauterbach TRACE32, Polarion

Thales – Gennevilliers

<May 2022 – July 2023>

Fields of expertise: Application layer development

Intervention title: LPM – Mission Preparation, Configuration & Maintenance Software

- **Goals and achievements:**

- ◆ Configuration management scripts
- ◆ Updating the application to match changes
- ◆ Bug fixes
- ◆ Documentation

- **Technical fields:**

- ◆ Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark

Valeo – Creteil

<March 2022 – March 2022>

Fields of expertise: Audit of tests for Valeo **Egypt**

Intervention title: All tests process audit and analysis

Arkamys – Levallois-Perret

<July 2021 – Feb. 2022>

Fields of expertise: Gstreamer framework

Intervention title: Audio plugins, unit tests, continuous integration & End to End tests

Project « Advanced audio car system »

▪ **Goals and achievements:**

- ◆ Plugins development in C language
- ◆ Unit tests, integration tests & End to End tests definitions and development
- ◆ Diagnostics

▪ **Technical fields :**

- ◆ Embedded Linux, Yocto, Buildroot, Valgrind, C Language, Gstreamer, CAN Elm327, Bash, Python, Awk, Sockets, Boundary devices IMX8, Agile, Jira

Thales – Vélizy-Villacoublay

<Nov. 2020 – June 2021>

Fields of expertise: Linux socket programming

Intervention title: IPC communication between different modules

Project « ATO – Autonomous train operation »

▪ **Goals and achievements:**

- ◆ Application Migration from 32Bit to 64Bit
- ◆ Linux socket programming
- ◆ Protocols and Packets definitions
- ◆ Diagnostics of network communications
- ◆ *IVV - Integration testing and validation*

▪ **Technical fields:**

- ◆ Embedded linux, C Language, UDP, TCP, Wireshark, Tshark, Pscapy, PyShark, Python, Awk, Netcat, Agile, Jira

PK Vitality – Paris

<June 2020 – Oct. 2020>

Fields of expertise: drivers and application development

Intervention title: Application development

Project « K'Watch Glucose CGM (Continuous Glucose Monitor) »

- **Goals and achievements:**

- ◆ HAL for the AFE (Analog Front End AD5940)
- ◆ PMIC supply Management (DA9070)
- ◆ BIAS injection
- ◆ Qspi driver and LOG management for the patient data
- ◆ Bluetooth Low energy BLE application With NORDIC NRF52840
- ◆ Touch Screen display Manager with STM32 TouchGFX, CubeMx and Touch Designer
- ◆ FreeRtos Tasks Management
- ◆ Low power mode management
- ◆ In Vitro tests of the Sensor

- **Technical fields :**

- ◆ STM32L4S9, NORDIC NRF52840, C, C++ Languages

ZODIAC AEROSPACE (SAFRAN) – Montreuil

<Feb. 2017 – March 2020>

Fields of expertise: Low level drivers and diagnostics

Intervention title: Software updates according to the new or modified requirements for the RCCB HVDC module

Project « AIRBUS GENOME »

- **Goals and achievements:**

- ◆ Optimization of low level layers (HAL) : CAN drivers
- ◆ Adding new features as Thermal memory implementation
- ◆ RS485 protocol definition and development
- ◆ Remote application monitoring
- ◆ *Unit testing* and validations
- ◆ Documentation according to the internal formats and standards

- **Technical fields :**

- ◆ Microcontroller Microchip DSPIC33, Mplab, C Language

Intervention title : uAFDX to CAN gateway development

Projet « AIRBUS CORAC »

- **Goals and achievements :**

- ◆ uAFDX to CAN interface definition and development
- ◆ Diagnostics
- ◆ *Unit testing* and validation

- **Technical fields :**

- ◆ Microcontroller DUAL Core TI Concerto, CCS Studio, C Language

Intervention title : Application development

Project «AIRBUS SSPC A350 AC9»

▪ **Goals and achievements :**

- ◆ Low level drivers optimisation
- ◆ development of the functions according to the specifications
- ◆ Unit testing and validation at VANCOUVER CANADA
- ◆ Documentation according to the internal formats and standards

▪ **Technical fields :**

- ◆ Microcontroller DUAL Core TI Concerto, CCS Studio, C Language

ENEDIS – Nanterre

<May 2015 – Sep. 2016>

Fields of expertise: diagnosing, bugs tracking and automating processes

Intervention title: Concentrator diagnosis

Project “Linky smart meters (smart grids)”

Goals and achievements:

- ◆ Diagnose concentrator
- ◆ Track concentrator bugs, i.e. creating, following up and validating bug fixes
- ◆ Audit and review coding with constructors
- ◆ Automate concentrator updating process
- ◆ Automate log recovering process
- ◆ Automate log analysis process and reports generation

Technical fields:

- ◆ Arduino, Raspberry PI, SimpleCV, Embedded linux, Kali linux, Cygwin, Bash, Perl, Awk, Python, Sonar, PC Lint

Sorin Medical – Clamart

<Nov. 2014 – April 2015>

Fields of expertise: Low level drivers, Low energy Bluetooth (Smart), Android

Intervention title: Feasibility study of the integration of a Bluetooth module (BLE) in a **Pacemaker**

Project “Platinum”

Goals and achievements:

- ◆ Validate choice of Bluetooth Smart module
- ◆ Develop the Serial Port Service Profile Auditing
- ◆ Conceive and develop communication protocol with the Smartphone
- ◆ Develop the Android application with Eclipse

- ◆ Conceive the hardware communication interface with the **implant**
- ◆ Prepare the implantation samples for the rabbits
- ◆ *Unit testing*

Technical fields:

- ◆ Cortex M0 microcontroller, KEIL, C Language, Android

General Electric Healthcare (GE Healthcare) – Buc

<Nov. 2012 – July 2014>

Fields of expertise: Low level drivers, integration, **Unit testing** and validation

Intervention title: Integrating CANOpen stack on RCB and CORTYX chips

Project “ZYX – Generator”

Goals and achievements:

- ◆ Integrate the CANOpen stack (slave) on the Rotor Control Board (RCB) chip
- ◆ Add new features to the BSP (Board Support Package) on the CORTYX chip based on the ARM Cortex A8 AM3505 processor
- ◆ Develop CAN drive for the CORTYX chip
- ◆ Develop CAN Service for the CORTYX chip
- ◆ Integrate the CANOpen battery (master) on the CORTYX chip
- ◆ Review code for TIF chip (Tube interface)
- ◆ Develop file management section on TIF Chip
- ◆ Develop TCL scripts for tests and validation of CAN with JSF (Homemade CAN tool)
- ◆ Write documents on achieved development and tests made
- ◆ Offer technical support for the US team and visits on the US site if needed
- ◆ Porting of CMX-RTX code on the DJINN chip
- ◆ Unit testing

Technical fields:

- ◆ TI TMS320F28335 microcontroller, XE167FH-200F, ARM Cortex A8 AM3505, VxWorks, C, C++ language

Gimelec – Algeria

<April 2012 – August 2012>

Fields of expertise: Low level drivers, RS485 communication, application

Intervention title: Finished product development

Project “Generator control unit”

Goals and achievements:

- ◆ Analyse needs and choices for technical solutions
- ◆ Develop electronic chip with μ Controller PIC16F628
- ◆ Develop communication drivers RS485
- ◆ Develop embedded application
- ◆ *Unit testing* and validation
- ◆ Develop setting tools on PC
- ◆ Write installation guide and user guide

Technical fields:

- ◆ PIC16F628 microcontroller, CCS PICC IDE, C language

PHOTOMATON – Paris

<May 2011 – Oct. 2011>

Fields of expertise: Low level drivers and application

Intervention title: Develop a payment system

Project “Payment system through coin-operated machine”

Goals and achievements:

- ◆ Analyse needs and choices for technical solutions
- ◆ Develop electronic board with μ Controller NXP LPC1754
- ◆ Develop crypto drivers Memory, IIC, LCD, Keyboard
- ◆ Integrate private use core and USB device stack (Virtual Comm)
- ◆ Set up boot loader and develop embedded application
- ◆ Lead function tests and validation
- ◆ Develop setting tools on PC
- ◆ Write installation guide and user guide
- ◆ Set up assembly line
- ◆ Choose subcontractors
- ◆ Launch a series of 10 products for website testing.

Technical fields:

- ◆ NXP LPC1754 microcontroller, IAR IDE, C language, Visual C#, Eagle (Cadsoft), Solidworks

BCM – Paris

<Jan. 2011 – April 2011>

Fields of expertise: Low level drivers and application

Intervention title: Develop a payment system

Project “Payment system through cards Mifare RFID”

Goals and achievements:

- ◆ Develop electronic board with μ Controller NXP LPC1754
- ◆ Develop drivers for Protocol ISO/IEC 14443-3 13.56 MHz contactless smart card standard
- ◆ Integrate USB device stack (CCID) and develop embedded application
- ◆ Unit testing
- ◆ Develop management tools for computer online wallet
- ◆ Write installation guide and user guide

Technical fields:

- ◆ NXP LPC1754 microcontroller, IAR IDE, C language, Visual C#, Eagle (Cadsoft)

Vertical M2M – Paris

<June 2010 – Nov. 2010>

Fields of expertise: Low level drivers and application

Intervention title: Develop a remote-read meter system

Project “Multi-use remote-read meter system (GSM/GPRS Modem)”

Goals and achievements:

- ◆ Analyse needs and choices for technical solutions
- ◆ Develop electronic board with μ Controller NXP LPC1763
- ◆ Develop GSM, SMS, Radio Frequency 433MHz low level drivers
- ◆ Set up boot loader and develop embedded application
- ◆ Unit testing and validation
- ◆ Develop setting tools on PC
- ◆ Design plastic box
- ◆ Write installation guide and user guide
- ◆ Set up assembly line
- ◆ Choose subcontractors
- ◆ Launch a series of 10 products for on field testing.

Technical fields:

- ◆ NXP LPC1754 microcontroller, IAR IDE, C language, Visual C#, Eagle (Cadsoft), Solidworks

PGS – Lyon

<March 2010 – May 2010>

Fields of expertise: Low level drivers and application

Intervention title: Develop a Bluetooth communication system

Project “Copier management system (classic Bluetooth)”

Goals and achievements:

- ◆ Analyze needs and choices for technical solutions
- ◆ Develop electronic board with μ Controller NXP LPC1343
- ◆ Develop Bluetooth drivers and USB Device (Virtual Comm)
- ◆ Develop embedded application
- ◆ Develop management and control application on computer
- ◆ Unit testing and validation

Technical fields:

- ◆ NXP LPC1343 microcontroller, IAR IDE, C language, Eagle (Cadsoft)

SFR Team

<Sep. 2009 – Feb. 2010>

Fields of expertise: Low level drivers and application

Intervention title: Develop a remote-read meter system

Project “Autonomous remote-read meter for LPG tank (battery driven, Modem GSM/GPRS)”

Goals and achievements:

- ◆ Analyze needs and choices for technical solutions
- ◆ Develop electronic chip with μ Controller PIC24F16KA102
- ◆ Develop GSM and SMS drivers
- ◆ Develop embedded application
- ◆ Unit testing and validation

Technical fields:

- ◆ MICROCHIP PIC24F16KA102 microcontroller, C language, CCS PICC IDE, Eagle (Cadsoft)

CARTADIS – Fontenay sous Bois

<Oct. 2001 – June 2009>

Fields of expertise: Low level drivers, middleware and application

Intervention title: Access control terminal and payment system

Project “Develop embedded low level drivers, embedded applications and high level applications on computer”

Goals and achievements:

- ◆ **TC4N project:** Control and magnetic card payment terminal
 - 1 Develop drivers (Assembly and C language) for Keyboard, Viewer, EEPROM, Reading and writing of the magnetic cards, RS232, RS485
 - 2 Develop embedded application with the smart and interactive menu manager
 - 3 Develop administration tools for cards under Visual Basic

Field: uPD70F3003 (V853) NEC, Green Hills Multi, C, Assembly Language, Visual Basic

- ◆ **TCP2 project:** Control and chip card payment terminal
 - 1 Develop chip card reading and writing drivers
 - 2 Develop embedded application featuring smart and interactive menu management
 - 3 Develop card administration tools under Visual basic

Environment: Hitachi H8S/214X, HEW IDE, C language, Visual Basic

- ◆ **TCBANK project:** Control and bank card payment terminal
 - 1 Develop bank card track reading and writing drivers
 - 2 Develop scripted communication protocols (DES)
 - 3 Develop server application (JAVA)

Environment: uPD70F3003 (V853) NEC, Green Hills Multi, C language, Net Beans

- ◆ **TCM2 project:** Contactless card reader (HID PROX 125 KHz)
 - 1 Develop card reading drivers
 - 2 Integrate USB HID (Human Interface Device) stack in Keyboard mode
 - 3 Develop administration tool

Environment: NXP LPC2148, IAR IDE, C language, Visual C#

- ◆ **P200N project:** Portable printer
 - 1 Develop drivers for thermal head
 - 2 Develop font creation tools

Environment: Hitachi H8S/214X, HEW IDE, C language, Visual studio

- ◆ **CMI project:** Smart network terminal
 - 1 Develop smart Switch drivers, SPI, RS485, RS232
 - 2 Develop drivers for graphic screen
 - 3 Develop communication protocols with RC4 scripting
 - 4 Integrate Tcpip, USB host, USB device, File system, Data Flash stacks
 - 5 Develop embedded server application
 - 6 Develop client application on computer

Environment: NXP LPC2468, IAR IDE, UCOS II, C language, Visual C#

- ◆ **TOTAL/ELF Cleaning project:** chip card payment solution for car cleaning
 - 1 Set up cross compilation tool chain (Embedded Linux)
 - 2 Develop drivers for chip cards, sensitive keyboards and graphic screen
 - 3 Develop application

Environment: ATMEL AT91RM9200, Build root, C language

- ◆ **TDA3 project:** Payment terminal for automatic teller machine
 - 1 Develop card reading drivers
 - 2 Develop drivers for EXE, MDB, BDV protocols

- 3 Develop double emulation interface application
- 4 Develop administration tools

Environment: Hitachi H8S/214X, HEW IDE, C language, Visual C#

Datus – SSII (Software Engineering Company)

Position: Embedded software engineer

<June 2001 – August 2001>

Fields of expertise: Low level drivers and application

Intervention title: Intern

Project “Motor control via PC (TGV model SNCF)”

Goals and achievements:

- ◆ Design piloting electronic board and power board
- ◆ Develop drivers and communication protocols
- ◆ Develop embedded application
- ◆ Develop control application on computer

Technical fields:

- ◆ MICROCHIP PIC16F873, MPLAB, C language, Visual Studio

Education

- ◆ **2001** Training Engineer of electronic and computer systems design, AFPA Champs-sur-Marne, France
- ◆ **1997** Diploma in Biomedical instrumentation, Tenon Hospital Paris
- ◆ **1994** Diploma in electronic engineering , communication and telecommunication , Boumerdes, Algeria

Foreign languages

- ◆ **English** professional proficiency
- ◆ **French**
- ◆ **Arabic** fluent